DGCA CPL / ATPL STUDY NOTES • RADIO NAVIGATION • CH 17

✈ Chapter 17: EFIS

Electronic Flight Information System

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1. Introduction

The Electronic Flight Information System (EFIS) replaces conventional electromechanical flight instruments with colour electronic displays. The EHSI (Electronic Horizontal Situation Indicator) displays navigation, radar and TCAS information in multiple selectable modes. EASA AMJ CS-25 governs colour coding and display requirements.

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Fig 17.1: EFIS overview — EADI/PFD (left) and EHSI/ND (right) with symbol generator inputs

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EHSI Inputs

- VOR (frequency, radial, deviation)
- DME (distance, groundspeed)
- ILS (localizer and glideslope deviation)
- ADF (relative bearing)
- FMS/FMC (waypoints, track, flight plan route)
- AWR (weather radar overlay)
- TCAS (traffic advisories)

All inputs are processed by **Symbol Generators** which output the display format.

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2. EFIS Components

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Fig: EFIS architecture — dual symbol generators feeding port and starboard EHSI and EADI displays

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EFIS System Architecture

Component	Function
EADI / PFD	Electronic Attitude Director Indicator / Primary Flight Display — attitude, airspeed, altitude, VSI, ILS deviations
EHSI / ND	Electronic HSI / Navigation Display — heading, track, bearing pointers, waypoints, weather
Symbol Generator	Processes all sensor inputs into display symbols for EHSI and EADI
EFIS Controller	Pilot interface — mode selection, range, data overlays (WXR, VOR, ARPT etc.)
EICAS/ECAM	Engine/systems monitoring display (Boeing/Airbus respectively)

3. EHSI Controller

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Introduction

(EASA CS-25) AMJ 25-11, contains the advisory material for manufacturers to observe when designing electronic horizontal situation indicator (EHSI) displays. It specifies the colour coding to be used, and the requirement on manufacturers to ensure there can be no confusion between colours or symbols. It also defines the probabilities that essential information (e.g. attitude, altitude, heading etc.) will not be lost or inaccurately displayed.

Detailed knowledge of the EASA CS-25 specifications is not required for the examination. Such knowledge as is needed has been reproduced in this chapter.

EHSI Controller

The EHSI displays navigational information, radar information and TCAS information. For the Radio Navigation examination knowledge of, and the ability to interpret, the navigational information is essential.

The inputs to the EHSI are from:

- IRS
- FMC
- VOR, DME, ILS, and ADF
- TCAS
- AWR

The information from all of the inputs is fed to the port and starboard EHSI, through the respective symbol generators, which are the heart of the EFIS. They process the various inputs to generate the correct symbology for the EHSI.

The EHSI controller has a function switch to select the mode of the displays, a range selection switch and 6 switches to control the display of data.

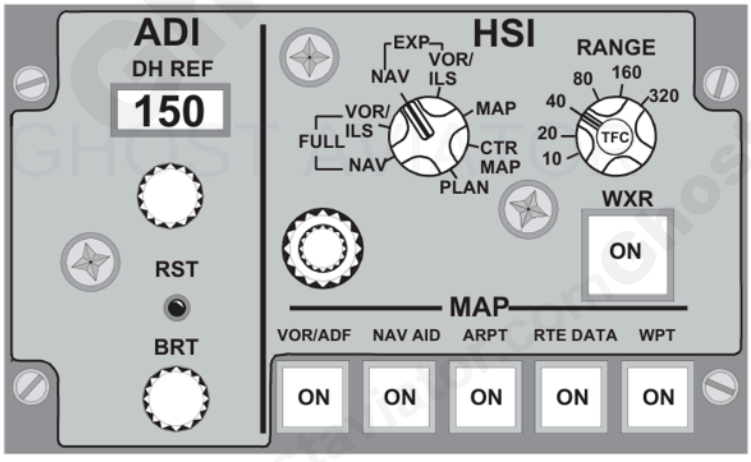


Figure 17.1

Fig: EFIS controller panel — mode selector, range knob, overlay switches (WXR, VOR, ARPT, WPT, etc.)

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EHSI Controller Functions

Control	Options / Effect
Mode Selector	ILS, VOR, MAP, PLAN (selects display format)
Range Selector	Selects scale of display (10, 20, 40, 80, 160, 320 NM)
WXR switch	Enables weather radar overlay (Expanded VOR/ILS and MAP modes only)
STA/VOR	Display tuned VOR/ADF radials
NAV AID	Display navigation aids (VOR, NDB symbols) from database
ARPT	Display airports from database
WPT	Display waypoints on MAP mode
RTE DATA	Display route data (ETAs, fuel predictions) at waypoints
TCAS switch	Toggle TCAS traffic display on/off; press to clear TCAS FAIL message

4. Display Modes

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The display modes available are:

- Full (or full rose) VOR/ILS.
- Expanded VOR/ILS.
- MAP.
- PLAN.

Weather radar and TCAS information can only be displayed on the expanded VOR/ILS and MAP displays. The selectable map background options are enabled in the Map and Plan modes.

The information available for display is:

- Tuned VOR/ADF radials (VOR/ADF).
- Navigation Aids (NAV AID).
- Airports (ARPT).
- Route Data (RTE DATA).
- Waypoints (WPT).
- Weather (WXR).

The traffic switch in the centre of the range selection knob when pressed will either:

- Display TCAS information, if not already displayed.
- Remove TCAS information from the display.
- When TCAS FAIL is displayed; remove the message.

With the exception of the PLAN mode which is orientated to true north, all the displays are orientated to aircraft heading which may be either magnetic or true. Range arcs (white) are displayed in the expanded VOR and ILS modes when the WXR switch is on, and in the MAP mode at all times.

EHSI mode options — Full Rose VOR, Expanded VOR, Full Rose ILS, Expanded ILS, MAP, PLAN

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EHSI Display Modes

Mode	Description	WXR/TCAS	Orientation
Full Rose VOR	360° compass rose; VOR course/deviation; two bearing pointers	✗	Heading-up
Expanded VOR	Arc display (~120° arc); VOR course, DME, deviation bar; range arcs shown	✓	Heading-up
Full Rose ILS	360° rose; ILS course/localizer deviation; glideslope on EADI	✗	Heading-up
Expanded ILS	Arc display; ILS/LOC course deviation; range arcs shown	✓	Heading-up
MAP	Moving map centered on aircraft; waypoints, airports, route, weather	✓	Heading-up
PLAN	Static map of flight plan; waypoints, airports, NAVAIDs; NOT heading-up	✗	True **North-up**

⚠ Key Points

- WXR and TCAS: only displayable in **Expanded VOR/ILS** and **MAP** modes
- PLAN mode is **North-up**; all others are **Heading-up** (magnetic or true)
- Range arcs (white): displayed in expanded modes when WXR on, and MAP at all times
- MAP mode selectable overlays: VOR/ADF, NAV AID, ARPT, RTE DATA, WPT, WXR, TCAS

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Display Modes

Expanded VOR Mode

The expanded VOR mode displays VOR information with a VOR selected and either a manual or database generated input of track.



Figure 17.2 Expanded VOR mode

The display shows that VOR2 is in use on a frequency of 116.80 MHz, the aircraft is outbound from the beacon at a range of 13.3 NM (DME) and is 7.5° right of the required track (165°(M)). The heading is 130°(M) and the present track 133°(M). The pilot has selected the heading bug to 104°(M). WXR is selected and the radar is in WX+T mode with 12° uptilt and the display is showing a contouring cloud centred on 105°(M) between 8 and 17 NM. The selected scale is 20 NM and the wind is 126°(M)/20 kt.

Full Rose VOR Mode



Figure 17.3 Full rose VOR mode

Fig 17.2: Expanded VOR mode — aircraft position relative to VOR, DME distance, heading, track, wind

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The full rose VOR mode is showing the same information as *Figure 17.4*, with some differences in symbology and the TO/FROM indication is a pointer that will appear either above or below the lateral deviation scale.

Expanded ILS Mode



Figure 17.4 Expanded ILS mode

The expanded ILS mode shows the appropriate ILS information when an ILS localizer frequency is selected. The glide slope indications are suppressed when the aircraft track is more than 90° removed from the ILS localizer course.

Full Rose ILS Mode



Figure 17.5 Full rose ILS mode

The full rose ILS mode shows the same information as the expanded ILS mode and has the same differences from the expanded ILS mode as noted for the VOR modes, except that the localizer deviation scale is doubled.

Fig 17.3: Full Rose VOR mode — bearing pointers, VOR1/VOR2 select, CDI deviation

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Map Mode



Figure 17.6 Map mode

The map mode shows the navigational information selected on the control panel and is heading orientated.

Plan Mode

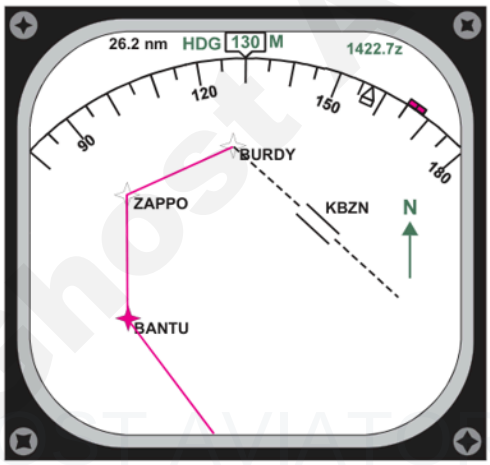


Figure 17.7 Plan mode

The plan mode is orientated to true north and the information displayed at the top of the screen is the same as in the map mode. The plan mode allows the pilot to review the planned route using the FMC/CDU LEGS page. The display will be centred using this page.

Fig: Expanded ILS mode — LOC deviation bar, range arcs, weather overlay, heading bug

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EHSI Colour Coding

The colour coding used on the EHSI is to the ICAO standard, which is also the standard adopted by EASA. The EASA CS-25 recommended colour presentation is:

Display features should be colour coded as follows:

Warnings	Red (R)
Flight envelopes and system limits	Red
Cautions, abnormal sources	Amber/Yellow (A)
Earth	Tan/Brown
Engaged modes	Green (G)
Sky	Cyan/Blue (C)
ILS deviation pointer	Magenta (M)
Flight director bar	Magenta/Green

Specified display features should be allocated colours from one of the following colour sets:

	Colour Set 1
Fixed reference symbols	White (W)
Current data, values	White
Armed modes	White
Selected data, values	Green
Selected heading	Magenta*
Active route/flight plan	Magenta
* Magenta is intended to be associated with those analogue parameters that constitute 'fly to' or 'keep centred' type information	

Precipitation and turbulence areas should be coded as follows:

Precipitation	0 - 1	mm/h	Black
	1 - 4	mm/h	Green
	4 - 12	mm/h	Amber/Yellow
	12 - 50	mm/h	Red
	Above 50	mm/h	Magenta
Turbulence			White or Magenta

Fig: MAP mode — waypoints, airways, airports overlaid on moving map with range rings

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EHSI Symbology

The symbology used in the B737-800 is depicted in the following table, which should be used in conjunction with the displays shown in *Figures* at *17.2* to *17.7.* Detailed knowledge of these symbols is not required for the EASA ATPL examinations.

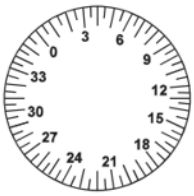
Symbol	Name	Applicable Modes	Remarks
200 NM or DME 124	Distance Display (W)	ALL	Distance is displayed to next FMC Waypoint (NM) or tuned Navaid (DME). Below 100 NM tenths of a NM will be displayed
HDG 263 M	HEADING Orientation (G) Indicator (W) Reference (G)	ALL	Indicates number under pointer is a heading - box indicates actual heading. Referenced to Magnetic (M) or true (TRU) North.
0835.4z	ETA Display (W)	PLAN, MAP	Indicates FMC computed ETA for the active waypoint.
	Expanded Compass Rose (W)	PLAN, MAP VOR, ILS	Compass Data is provided by the selected IRS (360° available but approximately 70° are displayed
	Full Compass Rose (W)	Full VOR, Full ILS	Compass Data is provided by the selected IRS.
	Aeroplane Symbol (W)	EXP VOR/ILS, MAP, PLAN	Represents the aeroplane and indicates its position at the apex of the triangle.
	Aeroplane Symbol (W)	Full VOR/ILS	Represents the aeroplane and indicates its position at the centre of the symbol.
	Waypoint Active (M) Downpath(W)	MAP, PLAN	Active - Represents the waypoint the aircraft is currently navigating to. Downpath - Represents a navigation point making up the selected active route.

Fig: PLAN mode — north-up static flight plan display; aircraft position shown as symbol

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5. EHSI Colour Coding (EASA CS-25)

Standard EHSI Colours

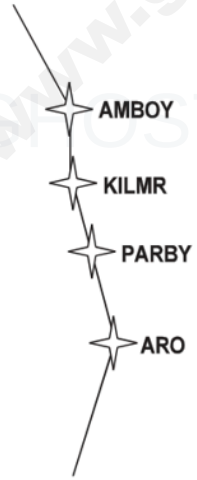
Colour	Used For
YELLOW/AMBER	Caution/advisory; items requiring attention but not immediate action
RED	Warnings; immediate action required; flight envelope exceedances
GREEN	Normal operation; active armed mode annunciation; on-course deviation
WHITE	Present values, scales, range arcs, heading information
CYAN/BLUE	Sky on EADI; engaged modes; selected altitude above present
MAGENTA	Active flight plan path; FMS/LNAV command bars; active waypoints

⚠ Colour Rules (CS-25)

- **RED** used only for warnings — never for advisory or normal information
- **AMBER/YELLOW** for cautions and non-normal conditions only
- System must ensure no confusion between colours or symbols is possible
- Critical information (attitude, altitude, heading) must not be lost or inaccurately displayed

6. EHSI Symbology

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	Trend Vector	MAP	Predicts aeroplane directional trend at the end of 30, 60 and 90 second intervals. Based on bank angle and ground speed. Three segments are displayed when selected range scale is greater than 20 NM, two on the 20 NM and one segment when on the 10 NM scale.
	Active Route (M) Active Route Mods (W) Inactive Route (C)	MAP, PLAN	The active route is displayed with continuous lines (M) between waypoints. Active route modifications are displayed with short dashes (W) between waypoints. When a change is activated in the FMC, the short dashes are replaced by a continuous line. Inactive routes are displayed with long dashes (C) between waypoints.
	Vertical Pointer (M) and Deviation Scale (W)	MAP	Displays vertical deviation from selected vertical profile (pointer) in MAP mode during descent only. Scale indicates +/- 400 ft deviation.
	Glide slope Pointer (M) and Deviation Scale (W)	ILS	Displays glide slope position and deviation in ILS mode.
	Wind Speed and Direction (W)	MAP, VOR, ILS	Indicates wind speed in knots and wind direction with respect to the map display orientation and compass reference.
	North Pointer (G)	PLAN	Indicates map background is orientated and referenced to true north.

EHSI symbology guide — aircraft symbol, bearing pointers, range arcs, course bar, TCAS intruders

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📡 Key EHSI Symbols

Symbol	Meaning
⇧ (aircraft symbol)	Own aircraft position; heading shown by nose direction
Course deviation bar	Lateral deviation from selected course (1 dot = typically 2° VOR or 0.5° LOC)
VOR1 bearing pointer (single)	Bearing to selected VOR1
VOR2/ADF pointer (double)	Bearing to VOR2 or ADF
Heading bug (cyan)	Selected heading (MCP/FCU heading select)
Wind arrow & velocity	Current wind direction and speed
Range arc (white)	Distance markers from aircraft; scale-dependent
TCAS traffic (cyan/amber/red)	Intruder aircraft; proximity advisory (cyan), traffic advisory (amber), resolution advisory (red)

7. Interpreting EHSI Displays

✓ Reading an Expanded VOR Display (Typical Exam Q)

Example: VOR2 selected, 116.80 MHz. Aircraft outbound from beacon, 13.3 NM (DME). 7.5° right of required track 165°(M). Heading 130°(M), track 133°(M). Heading bug 104°(M). WXR on WX+T mode, 12° uptilt, cloud 105°(M) at 8–17 NM. Scale 20 NM. Wind 126°(M)/20 kt.

- Course deviation bar is 7.5° to the LEFT of aircraft symbol → aircraft is 7.5° RIGHT of track
- Bearing pointer points behind aircraft (outbound from VOR)
- Weather contour centred at 105°(M) — shown as coloured return from radar

8. EFIS Summary

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Parameter	Detail
EHSI inputs	VOR, DME, ILS, ADF, FMS, AWR, TCAS
Processor	Symbol generators (port and starboard)
Display modes	Full Rose VOR/ILS, Expanded VOR/ILS, MAP, PLAN
WXR/TCAS display	Expanded VOR/ILS and MAP modes only
PLAN mode orientation	North-up (true); all others: heading-up
Active route	Magenta
Warnings	Red only
Cautions	Amber/yellow
Normal/scales	White
Active modes	Green
FMS commands	Magenta
Standard	EASA CS-25 AMJ 25-11

9. Practice Questions

Q1. The orientation of the PLAN mode of an EHSI display is:

(a) magnetic north-up

(b) track-up

(c) true north-up

(d) heading-up

Answer: (c)
PLAN mode is oriented to **true north-up**. All other EHSI modes (VOR, ILS, MAP) are heading-up.

Q2. Weather radar information can only be displayed in:

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(a) MAP mode only

(b) expanded VOR/ILS and MAP modes

(c) all modes

(d) full rose and MAP modes

Answer: (b)
WXR (weather) and TCAS can only be displayed in **Expanded VOR/ILS** and **MAP** modes.

Q3. The colour used for the active FMS flight plan path on an EHSI is:

(a) green

(b) white

(c) magenta

(d) cyan

Answer: (c)
The active FMS flight plan path is shown in **magenta**. Green = normal/active modes. White = scales/present values. Cyan = sky/selected altitude/heading bug.

Q4. The EHSI symbol generator receives inputs from:

(a) VOR, ILS, DME only

(b) VOR, ILS, DME, ADF, FMS, AWR, TCAS

(c) FMS only

(d) ILS and VOR only

Answer: (b)
Symbol generators process inputs from: **VOR, DME, ILS, ADF, FMS, AWR, and TCAS**.

Q5. According to EASA CS-25, which colour is used exclusively for warnings:

(a) amber

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(b) green

(c) white

(d) red

Answer: (d)
Red is used exclusively for warnings requiring immediate action. Amber/yellow for cautions. Green for normal operations. White for scales.

Q6. Range arcs on the EHSI are displayed in:

(a) MAP mode only

(b) all modes at all times

(c) expanded VOR/ILS when WXR selected, and MAP mode always

(d) full rose modes only

Answer: (c)
Range arcs (white) displayed in **expanded VOR and ILS modes when WXR is ON**, and in **MAP mode at all times**.

Q7. On the EHSI Expanded VOR display, the aircraft is shown 7.5° to the right of track 165°. The course deviation bar will appear:

(a) 7.5° to the left of the aircraft symbol

(b) 7.5° to the right

(c) directly ahead

(d) behind the aircraft

Answer: (a)
If the aircraft is to the RIGHT of track, the course bar (track) is displaced to the **LEFT** of the aircraft symbol — same as a CDI where the needle deflects toward the selected course.

Q8. The TCAS switch on the EHSI controller, when pressed with TCAS already displayed:

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(a) increases TCAS range

(b) removes TCAS from display

(c) changes TCAS mode

(d) has no effect

> **Answer: (c)**
> Pressing TCAS when displayed **removes TCAS information**. When TCAS FAIL is displayed, pressing it removes the message.

Q9. Selectable overlays available in MAP mode include:

(a) VOR/ADF, airports, waypoints, weather

(b) ILS only

(c) weather only

(d) waypoints and ILS only

> **Answer: (c)**
> In MAP mode, the selectable overlays are: **VOR/ADF radials, NAVAIDs, airports, waypoints, route data, weather (WXR), and TCAS**.

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